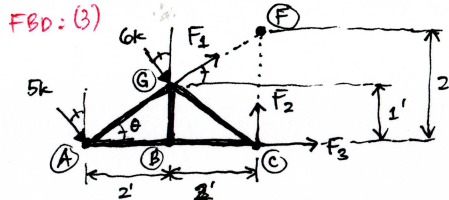


Problem 1

(a) Pass a section cut through FG, CF, and CD. Let F_1 , F_2 , and F_3 be the tensile forces in those members.

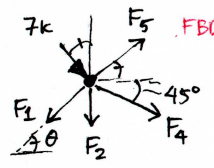


Observe that $\theta = \tan^{-1} 0.5$.

- Moment balance about F: $(5k) \sqrt{4^2 + 2^2} + (6k) \sqrt{2^2 + 1^2} + F_3(2) = 0 \Rightarrow F_3 = -8\sqrt{5} k \approx -17.889 k$
- Moment balance about C: $(F_1 \cos \theta)(1) + (F_1 \sin \theta)(2) + (6k)(\sin \theta)(1) = (6k)(\cos \theta)(2) + (5k)(\cos \theta)(4) \Rightarrow F_1 = 14.5 k$
- Moment balance about G: $(5k) \sqrt{2^2 + 1^2} + F_2(2) + F_3(1) = 0 \Rightarrow F_2 = 1.5\sqrt{5} k \approx 3.354 k$

∴ Forces in FG: ~~14.500 kN~~ 14.500 kN (T)
 CF: $1.5\sqrt{5} \approx 3.354$ kN (T)
 CD: $8\sqrt{5} \approx 17.889$ kN (C)

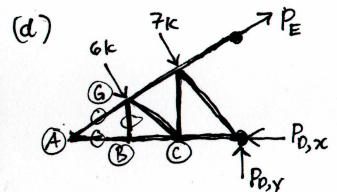
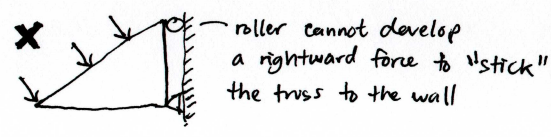
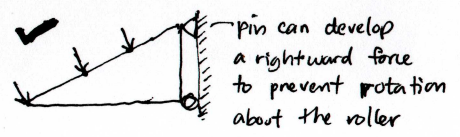
(b) Consider F and let F_4 and F_5 be the tensile forces in DF and EF, respectively.



- Force balance along horizontal and vertical directions:
- (2) $F_4(\cos 45^\circ) + F_5(\cos \theta) + (7k)(\sin \theta) = F_2(\cos \theta)$
- (2) $-F_4(\sin 45^\circ) + F_5(\sin \theta) = F_2(\sin \theta) + F_2 + (7k)(\cos \theta)$
- $\Rightarrow F_4 \approx -10.54092553 k ; F_5 = \frac{58}{3} k \approx 19.333 k$

∴ Forces in DF: 10.541 kN (C)
 EF: $\frac{58}{3} \approx 19.333$ kN (T)

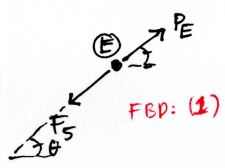
(c) The pin must be at E while the roller at D. The roller acts as a fulcrum and the pin will develop the counterforce. Should their locations be reversed, the pin will act as a fulcrum but the roller will not have the ability to develop a rightward counterforce.



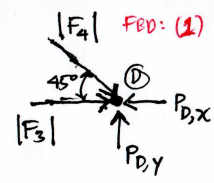
- AG, AB, and BG are obvious zero-force members
- case I case II
- Consequently, BG and BC satisfy case I
- ∴ Zero-force members: AG, AB, BG, BC

Problem 2, continued

(e) Consider E and realize that only two forces act on it: the tensile force in EF and the reaction developed in the pin. Therefore, equilibrium demands that the latter be of the same magnitude but opposite in direction to the former. So, the force developed in the pin at E is $\frac{58}{3} \approx 19.333$ kN directed $\theta = \tan^{-1} 0.5 \approx 26.565^\circ$ from the horizontal. [Equivalently, 17.292 kN (→) and 8.646 kN (↑).]

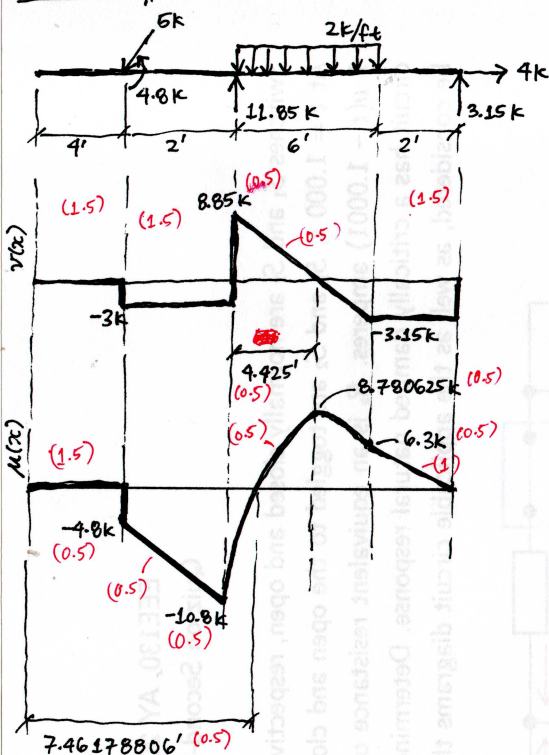


Consider D and realize that two compressive forces act on it (i.e., those developed in CD and DF) and that the reaction developed in the pin must counter their net. Equilibrium requires the horizontal component of the said reaction to be



- $|F_4|(\cos 45^\circ) + |F_5| \approx 25.342$ kN (←)
- and the vertical component to be
- $|F_4|(\sin 45^\circ) \approx 7.454$ kN (↑)
- [Equivalently, 26.415 kN directed 106.390° from the horizontal.]

Problem 2



(a) • moment balance about pin:

$$R_y(8) = \frac{3}{5}(5)(10) + 4.8 + 2(6)(5)$$

$$\Rightarrow R_y = 11.85 \text{ kip } (\uparrow) \quad (2)$$

• balance of horizontal forces:

$$P_x = \frac{4}{5}(5) \Rightarrow P_x = 4 \text{ kip } (\rightarrow) \quad (2)$$

horizontal reaction in pin

• balance of vertical forces:

$$P_y + R_y = \frac{3}{5}(5) + 2(6)$$

$$\Rightarrow P_y = 3.15 \text{ kip } (\uparrow) \quad (2)$$

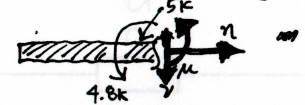
~~vertical reaction in pin~~

(b)-(d) Since there is a concentrated load at $x=4$, pass cuts just to the left and just to the right thereof. In both cases, we consider the left section of the beam for ~~case~~ ease.

At $x=4^-$ (just to the left of the load):

$$\begin{aligned} \cdot \eta &= 0 \text{ kip} \quad (0.5) \\ \cdot v &= 0 \text{ kip} \quad (1) \\ \cdot \mu &= 0 \text{ kip-ft} \quad (1.5) \end{aligned}$$

At $x=4^+$ (just to the right of the load):



• balance of horizontal forces:

$$\eta = \frac{4}{5}(5) \Rightarrow \eta = 4 \text{ kip} \quad (0.5)$$

• balance of vertical forces:

$$0 = v + \left(\frac{3}{5}\right)(5) \Rightarrow v = -3 \text{ kip} \quad (1)$$

• balance of moments about the section cut:

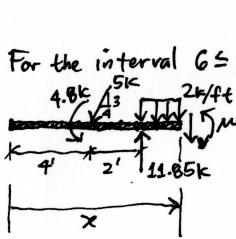
$$0 = \mu + 4.8 \Rightarrow \mu = -4.8 \text{ kip-ft} \quad (1.5)$$

Problem 2, continued

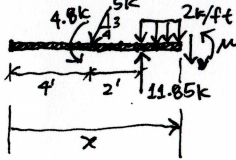
(e)-(f)

For the interval $0 \leq x < 4$:

$$\begin{aligned} \cdot v(x) &= 0 \text{ kip} \\ \cdot \mu(x) &= 0 \text{ kip-ft} \end{aligned}$$

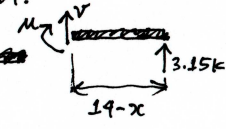


For the interval $6 \leq x < 12$:



$$\begin{aligned} \cdot 11.85 &= v + 3 + 2(x-6) \Rightarrow v(x) = -2x + 20.85 \text{ kip} \\ \cdot \mu + 4.8 + 3(x-4) + 2(x-6)\left(\frac{x-6}{2}\right) &= 11.85(x-6) \\ \Rightarrow \mu(x) &= -x^2 + 20.85x - 99.9 \text{ kip-ft} \end{aligned}$$

For the interval $12 \leq x < 14$:



$$\begin{aligned} \cdot v + 3.15 &= 0 \Rightarrow v(x) = -3.15 \text{ kip} \\ \cdot \mu &= 3.15(14-x) \Rightarrow \mu(x) = -3.15x + 44.1 \text{ kip-ft} \end{aligned}$$

Therefore,

$$v(x) = \begin{cases} 0 & \text{kip, } 0 \leq x < 4 \\ -3 & \text{kip, } 4 \leq x < 6 \\ -2x + 20.85 & \text{kip, } 6 \leq x < 12 \\ -3.15 & \text{kip, } 12 \leq x < 14 \end{cases} \quad (2)$$

$$\mu(x) = \begin{cases} 0 & \text{kip-ft, } 0 \leq x < 4 \\ -3x + 7.2 & \text{kip-ft, } 4 \leq x < 6 \\ -x^2 + 20.85x - 99.9 & \text{kip-ft, } 6 \leq x < 12 \\ -3.15x + 44.1 & \text{kip-ft, } 12 \leq x < 14 \end{cases} \quad (2)$$

(g)-(h) Observe that $v(x)$ changes sign from $x=6$ to $x=12$; i.e., it has a zero point when the expression for $6 \leq x < 12$ is zero:

$$-2\hat{x} + 20.85 = 0 \Rightarrow \hat{x} = 10.425' \text{ (i.e., } 4.425' \text{ to the right of the roller)}$$

Also, $\mu(x)$ has a vertex at $x=\hat{x}$ and therefore has a local maximum given by

$$-\hat{x}^2 + 20.85\hat{x} - 99.9 = 8.780625 \text{ kip-ft}$$

Lastly, observe that $\mu(x)$ changes sign from $x=6$ to $x=\hat{x}$, i.e., it has a zero point somewhere therebetween. Use the expression for $6 \leq x < 12$ and find the root thereof:

$$-\bar{x}^2 + 20.85\bar{x} - 99.9 = 0 \Rightarrow \bar{x} = 13.38821109', \text{ reject, out of bounds}$$

$$\Rightarrow \bar{x} = 7.46178806' \text{ (i.e., } 1.46178806' \text{ to the right of the roller)}$$

$$\Rightarrow \bar{x} = 7.46178806' \text{ (i.e., } 1.46178806' \text{ to the right of the roller)}$$